猪器官移植到人：2021-2026 年临床进展、关键风险与未来 3-5 年研究方向

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Contents

Abstract

猪到人异种移植在最近五年从长期前临床积累进入人体应用阶段。2021-2022 年脑死亡受者模型和 NEJM 报道的猪肾/猪心人体应用证明，经基因编辑的猪器官可以在人体循环中避免立即发生传统意义上的超急性排斥；2024-2026 年，活体猪肾异种移植、生理稳态追踪和免疫图谱研究，又把问题推进到可重复性、感染监管、免疫抑制强度、伦理选择和正式临床试验设计层面。目前更稳妥的判断是，异种移植尚未成熟，但临床可行性已经足以推动领域进入严格优化阶段。未来 3-5 年，猪肾最可能先进入可解释的早期临床试验；猪心仍需解决体量匹配、长期心肌适应、感染和排斥控制；猪肝/肺更适合先作为机器灌注、桥接支持或短期功能平台。研究重点应从单次突破转向连续病例安全性、标准化供体猪、跨物种感染监测、组织/液体活检整合和患者净获益评估。

1 引言

异种移植的临床动力来自供需矛盾。终末期肾病、终末期心衰和肝肺衰竭患者数量远超可用同种异体器官。猪作为供体来源具备繁殖、体量、器官解剖和基因工程可塑性优势，但跨物种屏障严苛：糖抗原、补体、凝血、炎症、先天免疫、T/B 细胞反应、病毒安全和伦理监管共同决定结果。过去十年，GGTA1、CMAH、B4GALNT2 等糖抗原敲除，人源补体/凝血调节蛋白表达，以及 PERV 处理、病原体净化和 CD40/CD154 轴免疫抑制，使前临床非人灵长类模型取得显著进展 [1–3]。

人体应用的突破改变了问题性质。2022 年 NEJM 报道的两例猪到人肾异种移植脑死亡模型，为观察人体免疫和生理反应提供了比非人灵长类更直接的证据 [4]。同年，首例基因编辑猪心移植到终末期心衰患者，受者生存约两个月，证明心脏异种移植可在同情使用场景下实现短期生命支持，同时也暴露感染、免疫抑制和长期适应问题 [5, 6]。2025 年 NEJM 报道猪肾用于终末期肾病活体受者，以及 Nature Communications / Nature Medicine 对活体猪肾受者的生理稳态和免疫图谱分析，标志着领域进入机制复盘和试验准备阶段 [7–9]。

2 猪肾：最接近正式临床路径的器官

猪肾的临床路径相对清晰。肾功能可通过尿量、肌酐、电解质、酸碱平衡和容量状态连续评估；移植物失败后患者理论上可回到透析，风险可逆性强于心脏或肝脏；肾移植也已有成熟的活检、DSA、dd-cfDNA 和免疫抑制监测框架。脑死亡模型和活体病例显示，基因编辑猪肾能够短期维持滤过和体液稳态，并为跨物种免疫图谱提供人体材料 [4, 8, 9]。

但猪肾临床化不能只看“接上后能工作”。更关键的问题包括早期抗体介导损伤、血栓性微血管病、炎症性内皮激活、补体逃逸是否持久、蛋白尿和肾小管功能是否稳定、受者是否产生影响未来同种异体移植的抗体，以及免疫抑制是否可在感染风险可接受的范围内长期维持 [10, 11]。脑死亡模型虽然宝贵，但脑死亡后的激素、炎症和血流动力学状态与活体患者不同，仍不能替代正式临床试验。

3 猪心：概念已被证明，长期成功仍未建立

猪心异种移植的临床意义巨大，因为终末期心衰患者可替代方案有限，机械循环支持并不等同于长期根治。2022 年首例基因编辑猪心移植证明，跨物种心脏可以在人体内短期承担生命支持功能 [5]。随后领域围绕体量匹配、连续非缺血保存、免疫抑制、PCMV/PRV 感染、心肌过度生长、微血栓和慢性血管病展开复盘 [12–14]。

与猪肾不同，猪心失败往往没有透析式的低风险回退路径。因此，猪心早期临床试验的伦理门槛更高，更适合在无其他生命延续选择、机械支持不可行或作为严格桥接方案的患者中谨慎推进。未来若要从同情使用走向试验，必须证明供体猪病原体控制、心脏保存和受者免疫方案能够带来可重复的中期生存，而不能依赖单例突破。

4 讨论：感染、安全与监管

感染风险是异种移植区别于同种异体移植的根本维度。PCMV/PRV 被认为与早期心脏异种移植结局密切相关，PERV 虽然可通过供体筛选和基因编辑降低风险，但其公共卫生含义要求长期监测受者、密切接触者和样本库 [12, 15]。正式试验需要明确供体猪指定病原体阴性标准、繁育设施质量、器官采集链路、术后病毒检测频率、阳性触发处置和长期随访责任。

伦理上，初始受试者选择必须避免把“无路可走”简单等同于“可接受任何风险”。候选者需要理解异种移植可能影响未来同种异体移植机会、需要长期感染监测、可能出现未知公共卫生风险，并可能承担高强度免疫抑制副作用 [16–18]。知情同意也不应只是一次签字，而应是包含教育、复核和独立伦理支持的连续过程。

5 机器灌注与异种移植的交叉机会

机器灌注可能成为异种移植从“能移植”走向“能筛选和修复”的关键平台。对猪肾、猪肝、猪肺和猪心，离体灌注可用于评估血管阻力、代谢恢复、补体激活、凝血倾向、内皮损伤、病毒污染和人血暴露后的免疫反应。近年“xenoperfusion”概念和 alpha-Gal-KO 猪肝/肺常温机器灌注研究提示，灌注平台可作为人体前的功能压力测试 [19, 20]。

这一路线对猪肝和猪肺尤其重要。肝脏涉及复杂合成、解毒、凝血和免疫功能，肺脏与血气屏障和炎症反应高度相关，直接长期植入人体风险更高。未来 3-5 年，猪肝/肺更现实的临床入口可能是短期桥接、离体灌注评估、人血灌注安全性和急性功能支持，而非立即追求长期植入。

6 展望：未来 3-5 年研究方向

1. 猪肾早期临床试验应采用连续小队列、哨兵病例和预定义停试规则。核心终点应包括患者生存、移植物功能、严重感染、ABMR/TMA、回到透析能力、生活质量和未来同种异体移植可及性。
2. 建立“供体猪最小可接受标准”。至少应覆盖三糖抗原敲除、人源补体/凝血调节、SLA/组织相容性策略、PERV/PCMV/PRV 检测、繁育环境、体量匹配和器官质控。

3. 发展跨物种液体活检。dd-cfDNA、猪源 cfDNA、EV、补体片段、凝血标志物、DSA/抗猪抗体和病毒核酸应形成组合监测，用于早期区分缺血损伤、ABMR、感染和凝血微血管病。
4. 用机器灌注做人血压力测试。移植前以受者血浆或标准化人血组分灌注，评估补体、凝血、内皮损伤和病毒释放，可帮助筛除高风险器官和优化个体化免疫方案。
5. 比较 CD40/CD154 阻断、补体抑制、B 细胞/浆细胞靶向和抗凝策略的最小有效组合。目标不是最大免疫抑制，而是可长期承受的免疫-感染平衡。
6. 建立长期公共卫生随访体系。异种移植试验的成功不仅是单个患者获益，还包括对潜在跨物种感染风险的透明监测、样本留存、数据共享和国际监管协调。
7. 将患者选择从"绝望程度"转向"净获益概率"。猪肾候选者可优先考虑透析负担极高、同种异体移植机会极低但感染风险可控者；猪心候选者则需更严格的替代方案评估。

7 结论

猪到人异种移植已经越过"是否可能"的第一道门槛，但尚未越过"是否可重复、可监管、可普及"的第二道门槛。猪肾最可能率先形成正式临床试验路径，因为失败后仍有透析回退，且监测框架成熟；猪心的临床意义同样重大，但风险承受空间更窄；猪肝和猪肺更适合先通过机器灌注和短期桥接建立安全边界。未来 3-5 年，领域更需要连续、透明、机制驱动的临床研究，把供体工程、免疫控制、感染监管和患者净获益放在同一个证据框架中。

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